

Ayush Maurya

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SUMMARY

Machine Learning Research Engineer experienced in experiment design, statistical analysis, and reproducible research prototyping with PyTorch and Python. Delivered a 92%-accuracy geolocation model through controlled hypothesis testing, an open-source agent-based simulation package on PyPI, and LLM-powered document recovery pipelines. Combines deep learning expertise with scientific computing and rigorous methodology for end-to-end research execution from hypothesis formation through deployment.

EDUCATION

IIIT Hyderabad	2021–Present
B.Tech in Computer Science and M.S. by Research in Computing and Human Sciences	
CHD: Computer Science and Humanities Integrated Dual Degree (Offered by the Human Sciences Research Center)	
Mary Gardiner's Convent School	2021
Class XII — 90.1%	
City Montessori School	2019
Class X — 94.2%	

EXPERIENCE

SERC Digital Twin	Environmental IoT
- Developed a real-time environmental monitoring dashboard using the MERN stack. - Integrated local sensors for temperature, humidity, water, and air quality monitoring.	
TAFEA	System Design
- Designed a student client software system using standard SDLC methodologies. - Integrated LLM APIs to support tailored student-facing features and workflows.	

PROJECTS

ElectoralSim	Agent-Based Electoral Simulation
- Published open-source research prototype for vectorized agent-based electoral simulation on PyPI. - Implemented modular simulation framework supporting multiple electoral systems with documented API.	
ML for Local Geoguesser	Visual Geolocation via Deep Learning
- Designed controlled experiments comparing ResNet50 architectures for visual geolocation inference. - Achieved 92% region accuracy through iterative hypothesis testing and statistical model evaluation.	
Highland History Project	OCR + LLM Pipeline
- Built OCR-LLM pipeline digitizing historical documents for Himalayan trade-network research. - Evaluated multiple LLM configurations for degraded-text recovery with accuracy benchmarks.	
Design for Social Innovation (TAFEA)	AI-Assisted Education Platform
- Developed AI-assisted platform enabling curriculum design for resource-constrained classrooms. - Conducted human-centered research and evaluated prototype effectiveness through user studies.	
Age Prediction Model	Transfer Learning vs Ensemble
- Conducted comparative study of transfer learning versus ensemble methods for age prediction. - Formulated experimental hypotheses, implemented in PyTorch, and documented statistical findings.	

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, SQL, HTML/CSS
Technologies: Machine Learning, PyTorch, Linux, System Design, Systems Programming
UI/UX and Design: Wireframing, Prototyping, User Stories, User Research, Interaction Design

HONOURS

UGEE Rank: 111 out of 40,000+ candidates

ACTIVITIES

Bass guitarist in college band and lawn tennis player.